Deep Learning Framework for Automated Plant Disease Detection Using Leaf Images

A Mini Project Report Submitted in partial fulfillment of the requirements for the award of the degree of

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In

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Or

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

Or

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

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The mini-project report entitled "Deep Learning Framework for Automated Plant Disease Detection Using Leaf Images" is a record of bona fide work carried out by the following students in partial fulfillment of the requirements for the award of the B.Tech degree at K L (Deemed to be) University.

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CERTIFICATE

This is to certify that the mini-project report entitled "Deep Learning Framework for Automated Plant Disease Detection Using Leaf Images" is a bona fide work carried out by the following students under the supervision of [Faculty Guide Name] during the academic year [20__–20__], in partial fulfillment of the requirements for the award of the Bachelor of Technology degree at K L (Deemed to be) University.

Signature of the Project Supervisor

Signature of the HOD **Signature of the External Examiner**

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TABLE OF CONTENTS

S.No.	Contents	Page no.
1.	Abstract	
2.	Introduction	
3.	Literature survey	
4.	Problem Statement and Objectives	
5.	Hardware and Software requirements	
6.	Implementation	
7.	Results & Discussion	
8.	Conclusion & Future Scope	
9.	References	

1. ABSTRACT

Agriculture plays a vital role in food security and economic development. However, plant diseases significantly affect crop yield and quality, causing financial losses to farmers. Traditional disease identification through manual inspection is time-consuming and requires expert knowledge. This project proposes a Deep Learning Framework for Automated Plant Disease Detection Using Leaf Images to identify plant diseases efficiently.

The proposed system uses the PlantVillage dataset, containing 54,306 images of healthy and diseased leaves across 14 crop species and 26 diseases. The PlantDoc dataset, containing real-world plant images, is also considered for evaluating generalization. Image preprocessing techniques such as resizing, normalization, and data augmentation are applied to improve the quality and diversity of training images.

A pre-trained EfficientNet model is used with Transfer Learning and Convolutional Neural Network techniques to extract image features and classify plant leaves into healthy and diseased categories. Unlike text analytics approaches, this project uses image-based feature extraction rather than TF-IDF or text embeddings.

The model is intended to be evaluated using Accuracy, Precision, Recall, F1-score, and a Confusion Matrix. A Streamlit-based interface is proposed to allow users to upload leaf images and receive disease predictions with confidence scores. The system aims to support early disease identification, reduce manual effort, and promote sustainable agricultural practices.

Team Name:

Deep Learning Framework for Automated Plant Disease Detection Using Leaf Images

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2. INTRODUCTION

Agriculture plays an important role in food security and economic development. However, plant diseases are one of the major challenges faced by farmers, as they affect crop quality, reduce productivity, and cause financial losses. Early identification of plant diseases is essential to prevent their spread and support effective crop management.

Traditional plant disease identification mainly depends on manual inspection by farmers or agricultural experts. Examining large numbers of plants manually is time-consuming, labour-intensive, and requires specialized knowledge. Furthermore, expert assistance may not always be available in remote agricultural areas. Deep learning and computer vision provide an opportunity to automate this process by identifying visual patterns in plant leaf images, such as spots, colour changes, and damaged regions.

The proposed project, **Deep Learning Framework for Automated Plant Disease Detection Using Leaf Images**, aims to develop an automated system that identifies whether a plant leaf is healthy or affected by a particular disease.

The intended users include farmers, agricultural researchers, and agricultural extension workers. The system accepts a plant leaf image as input and predicts the corresponding disease category along with a confidence score.

The proposed system follows a structured pipeline:

1. **Data Collection:** Collect labelled plant leaf images from the PlantVillage dataset and consider PlantDoc images for real-world evaluation.
2. **Image Preprocessing:** Resize and normalize images and apply data augmentation techniques such as rotation, flipping, and scaling.
3. **Feature Extraction:** Use a pre-trained EfficientNet model to extract important visual features from leaf images.
4. **Model Training:** Apply Convolutional Neural Networks (CNNs) and Transfer Learning to train and fine-tune the classification model.
5. **Model Evaluation:** Evaluate the model using Accuracy, Precision, Recall, F1-score, and a Confusion Matrix.
6. **User Interface:** Develop a Streamlit-based interface that allows users to upload leaf images and view disease predictions with confidence scores.

The remaining chapters of this report explain the proje

3. LITERATURE SURVEY

The following literature survey is adapted to your actual project, Deep Learning Framework for Automated Plant Disease Detection Using Leaf Images. Since your project uses image classification, the reviewed studies focus on computer vision, CNNs, transfer learning, and plant disease detection rather than text analytics.

2.1 Review of Existing Research Papers

1. Mohanty et al. (2016) – Using Deep Learning for Image-Based Plant Disease Detection

- Problem: Automated identification of plant diseases from leaf images.
- Dataset: PlantVillage, containing 54,306 images covering 14 crop species and 26 diseases.
- Method: AlexNet and GoogLeNet convolutional neural networks.
- Evaluation and findings: GoogLeNet achieved 99.35% accuracy on the held-out test set.
- Limitation: Performance dropped to 31.4% on images collected under different conditions, showing limited real-world generalization. [1]

DOI+1

2. Singh et al. (2020) – PlantDoc: A Dataset for Visual Plant Disease Detection

- Problem: The lack of real-world plant disease images limits practical disease detection systems.
- Dataset: PlantDoc, containing 2,598 images across 13 plant species and up to 17 disease classes.
- Method: The authors trained three models for plant disease classification.
- Evaluation and findings: Their experiments showed that using the PlantDoc dataset could improve classification accuracy by up to 31%.
- Limitation: The dataset is relatively small and contains internet-sourced images, which can introduce variations in quality and labelling. [2]

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3. Atila et al. (2021) – Plant Leaf Disease Classification Using EfficientNet Deep Learning Model

- Problem: Improving automated plant leaf disease classification.
- Dataset: PlantVillage, with 55,448 original images and 39 classes.
- Method: Compared EfficientNet architectures with other deep learning models using transfer learning.

- Evaluation and findings: EfficientNet-B4 and EfficientNet-B5 achieved the highest reported accuracies of 99.97% and 99.91%, respectively.
- Limitation: The reported dataset is primarily controlled, so the results do not alone establish performance under varied field conditions. [3]

ScienceDirect

4. Dhaka et al. (2021) – A Survey of Deep Convolutional Neural Networks Applied for Prediction of Plant Leaf Diseases

- Problem: Reviews deep-learning approaches for automatic plant leaf disease identification.
- Dataset: Examines commonly used datasets, including PlantVillage, and compares research involving different crops and diseases.
- Method: Reviews CNN architectures, preprocessing methods, optimization techniques, and evaluation metrics.
- Evaluation and findings: Reports that CNN-based methods have achieved strong classification performance, with some PlantVillage experiments reporting accuracy above 99%.
- Limitation: Highlights that performance depends on image acquisition conditions, dataset quality, and model selection. [4]

PMC

5. Plant Disease Identification on Real-World Data Using Deep Learning (2021)

- Problem: Evaluates whether deep-learning models can identify plant diseases in realistic images rather than only controlled laboratory images.
- Dataset: PlantDoc, containing images of different crops and disease classes.
- Method: Compares InceptionResNet, EfficientNetLite, and VGG-19 with different training settings, augmentation methods, and loss functions.
- Evaluation and findings: EfficientNetLite achieved 70.71% accuracy, a mean F1-score of 0.70, and 95.57% top-5 accuracy.
- Limitation: The relatively lower classification performance demonstrates the difficulty of handling complex backgrounds and field conditions. [5]

IEEE Xplore

6. Acharya et al. (2026) – Plant Disease Identification Through Explainable AI Using Fine-Tuned EfficientNet

- Problem: Improves plant disease classification while making model predictions more interpretable.
- Dataset: Used 20,639 PlantVillage images covering 15 classes and 4,432 drone-captured rice images.
- Method: Applied EfficientNetV2-B1 with progressive fine-tuning and Grad-CAM visualization.
- Evaluation and findings: Reported 99.7% accuracy on PlantVillage and 99.03% on the drone-based rice dataset.
- Limitation: The approach involves model fine-tuning and additional validation techniques, which can increase computational and implementation complexity.

4. PROBLEM STATEMENT AND OBJECTIVES

Problem Statement:

To develop a deep learning-based automated plant disease detection system using leaf images that can accurately classify healthy and diseased plants and provide disease predictions with confidence scores.

Problem Description:

Plant diseases affect crop quality and agricultural productivity, resulting in financial losses for farmers. Traditional disease identification relies on manual inspection and expert knowledge, making the process time-consuming and difficult to access in remote areas. This project proposes an EfficientNet-based deep learning framework using transfer learning to automate plant disease classification and support early disease identification. The intended users include farmers, agricultural researchers, and agricultural extension workers.

Objectives:

The main objectives of this project are:

- **Dataset Collection: Obtain and inspect labelled plant leaf image datasets, including PlantVillage and PlantDoc, and document their sources, image counts, crop categories, and disease classes.**
- **Image Preprocessing: Resize and normalize leaf images and apply suitable data augmentation techniques to improve image quality and dataset diversity.**
- **Feature Extraction: Use a pre-trained EfficientNet model to extract important visual features from plant leaf images.**
- **Model Development: Develop and train a CNN-based plant disease classification model using transfer learning and fine-tuning techniques.**
- **Model Evaluation: Evaluate the trained model using Accuracy, Precision, Recall, F1-score, and a Confusion Matrix.**
- **Error Analysis: Examine incorrect predictions and discuss challenges such as complex backgrounds, lighting variations, image quality, and class imbalance.**
- **User Interface Development: Develop a Streamlit-based interface that allows users to upload plant leaf images and view the predicted disease category with a confidence score.**
- **Practical Application: Design the system to support preliminary plant disease identification and promote efficient agricultural disease monitoring.**

5. HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements: A computer or laptop with a modern multi-core processor, minimum 8 GB RAM, sufficient storage, and internet access. A GPU is optional but recommended for faster model training.

Software Requirements: Python 3.x, Google Colab or Jupyter Notebook, TensorFlow/Keras or PyTorch, NumPy, Pandas, OpenCV, Matplotlib, and Scikit-learn. PlantVillage and PlantDoc datasets are used for training and testing. Streamlit is used for the user interface, and Git/GitHub for version control.

6. IMPLEMENTATION

1. Data and Task: The project uses PlantVillage and PlantDoc leaf image datasets to identify healthy and diseased plants. Images are classified into different disease categories. The dataset is divided into training, validation, and testing sets.

2. Image Preprocessing and Features: Images are resized, normalized, and augmented to improve model performance. Image features are extracted automatically using CNN and EfficientNet. Text preprocessing techniques such as tokenization and TF-IDF are not applicable.

3. Model Training and Evaluation: EfficientNet with transfer learning is used for training. Model performance is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix. The best model is selected based on validation performance.

A. Project Workflow and Modules

- Data loading: Load PlantVillage and PlantDoc datasets.
- Image preprocessing: Resize, normalize, and augment images.
- Model training: Train the EfficientNet-based deep learning model.
- Evaluation: Generate performance metrics and confusion matrix.

B. Deep Learning Implementation

The project uses CNN and EfficientNet for plant disease classification. Transfer learning helps improve feature extraction and reduce training time. Python libraries such as TensorFlow/Keras or PyTorch are used for model development.

C. Demonstration / Interface

A Streamlit interface allows users to upload a leaf image and view the predicted disease category with its confidence score.

7. RESULTS AND DISCUSSION

- The project results are evaluated using the PlantVillage and PlantDoc datasets. The analysis includes dataset distribution, image preprocessing, model performance, and disease classification results.

- The model performance is measured using accuracy, precision, recall, F1-score, and a confusion matrix. Class-wise performance is also analyzed to identify correctly and incorrectly classified images.

- Screenshots of the dataset, preprocessing results, training graphs, confusion matrix, and Streamlit prediction interface are included to demonstrate the system's performance. The results are discussed to identify classification errors, class imbalance, and limitations such as variations in lighting, background, and leaf image quality.

8. CONCLUSION AND FUTURE SCOPE

Conclusion

This project focuses on developing a deep learning-based system for automated plant disease detection using leaf images. CNN and EfficientNet with transfer learning are used to classify plant diseases using the PlantVillage and PlantDoc datasets. Model performance is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix. The project aims to simplify disease identification and provide predictions through a user-friendly interface. The final achievement of each objective and the measured model performance will be confirmed after implementation and testing. The major limitation is that variations in lighting, background, and image quality may affect prediction accuracy.

Future Scope

- Train and test the model using larger and more diverse real-world leaf image datasets.
- Improve performance on imbalanced disease classes.
- Use explainable AI techniques such as Grad-CAM to highlight affected leaf regions.
- Develop a mobile-friendly application for farmers.
- Support multiple languages in the user interface.
- Test model performance under different environmental conditions and measure prediction time.

9. REFERENCES

1. Hughes, D. P., & Salathé, M. (2015). *An open access repository of images on plant health to enable the development of mobile disease diagnostics*. arXiv. **https://arxiv.org/abs/1511.08060**
2. Singh, D., et al. (2020). *PlantDoc: A Dataset for Visual Plant Disease Detection*. ACM CoDS-COMAD, pp. 249–253. **https://doi.org/10.1145/3371158.3371196**
3. Mohanty, S. P., Hughes, D. P., & Salathé, M. (2016). *Using Deep Learning for Image-Based Plant Disease Detection*. Frontiers in Plant Science, 7, 1419. **https://doi.org/10.3389/fpls.2016.01419**
4. Tan, M., & Le, Q. V. (2019). *EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks*. Proceedings of the International Conference on Machine Learning (ICML). **https://proceedings.mlr.press/v97/tan19a.html**
5. Scikit-learn Documentation. **https://scikit-learn.org/stable/**
6. TensorFlow Documentation. **https://www.tensorflow.org/guide**
7. OpenCV Documentation. **https://docs.opencv.org/**
8. Streamlit Documentation. **https://docs.streamlit.io/**
9. PlantVillage Dataset Repository. **https://github.com/spMohanty/PlantVillage-Dataset**
10. PlantDoc Dataset Repository. **https://github.com/pratikkayal/PlantDoc-Dataset**